UCLA Samueli
Civil & Environmental Engineering

Distinguished TA Award Selection Committee
University of California, Los Angeles

Dear Members of the Distinguished TA Award Committee,

I am writing in my capacity as Vice Chair for Graduate Studies in the Civil and Environmental Engineering Department to offer my strong and unreserved support for Ms. Maryam Ghamarchehreh's nomination for the Distinguished TA Award at UCLA.

As part of our departmental review process, I have examined the teaching evaluations Maryam received over the past year. It is clear from these materials that students have responded very positively to her work as a teaching assistant. They consistently praise both her teaching effectiveness and her supportiveness. Representative comments include: "*Wonderful TA who really helped my learning*" and "*One of the best TAs I have ever had*." These statements are not isolated; they reflect a broader pattern of appreciation for Maryam's clarity, responsiveness, and dedication to her students' success.

In addition to her strong record of teaching, Maryam has also demonstrated notable professionalism and maturity in handling challenging situations. A recent incident involved a very disrespectful email she received from a student in her class. Maryam responded to this situation with admirable composure and judgment. She replied to the student in a respectful and measured way, while also appropriately escalating the issue to the relevant faculty and staff. Through this process, she ensured that the matter was handled fairly and smoothly resolved. Her response exemplified the professional conduct and emotional intelligence we seek to foster in our TAs.

Taken together—her consistently positive student evaluations, the high regard in which she is held by her students, and her capacity to navigate difficult interactions with professionalism—Maryam clearly stands out as an exemplary teaching assistant. She contributes not only to her students' learning, but also to the collegial and respectful climate of our department.

For these reasons, Maryam has the department's full and unreserved support for the Distinguished TA Award. I am confident that she is highly deserving of this recognition.

Thank you for your consideration.

Sincerely,

Sincerely,



Sriram Narasimhan, PhD, P.Eng (Ont), F.ASCE,

Professor and Graduate Vice Chair, C&EE

Self-Statement of Teaching Accomplishments

As a Teaching Assistant for undergraduate and graduate engineering courses, I have contributed significantly to student learning through effective instruction, fair and transparent assessment, and consistent student support. My teaching accomplishments reflect a strong commitment to clarity, professionalism, and creating a respectful learning environment.

One of my primary accomplishments is my ability to make technically challenging material accessible to students who come from a wide range of academic backgrounds. In the courses I teach, students often have very different levels of prior preparation and apply the material in diverse disciplinary contexts. In discussion sections and office hours, I explain complex concepts by breaking them into clear, logical steps and illustrating how the same methods and practices are applied across different fields. I deliberately adapt my explanations and examples so students from different majors can relate the material to their own academic experiences. This approach has helped students who initially struggled gain confidence and improve their performance on assignments and exams.

I have played a key role in assessment and feedback through careful and consistent grading. I grade assignments and exams using established rubrics and prioritize completing grading as quickly as possible so students have sufficient time to review feedback and learn from their mistakes before subsequent assessments. I provide detailed written comments that highlight both strengths and areas for improvement. When I identify inconsistencies or ambiguities in grading guidelines or provided solutions, I communicate directly with the course instructor to ensure fairness and consistency across all sections. In one instance, this process resulted in grade corrections for multiple students, reinforcing transparency, equity, and consistency in evaluation.

Another major teaching accomplishment is my responsiveness and availability to students. I maintain regular office hours and respond promptly to student emails, often clarifying course expectations, addressing confusion about assignments, or guiding students on effective problem-solving strategies. Many students who attend office hours repeatedly show noticeable improvement in both understanding and confidence.

My teaching contributions extend beyond academic instruction to include modeling professionalism and ethical conduct in the learning environment. I consistently demonstrate respectful and professional communication in all interactions with students, instructors, and staff, contributing to a positive and supportive classroom climate. I maintain clear boundaries while remaining approachable and responsive to student concerns. For example, when a student expressed dissatisfaction with a grade through an unprofessional email, I responded calmly and respectfully, focusing on course policies and constructive guidance. When similar behavior continued, I followed appropriate academic procedures by forwarding the full correspondence to the course instructor and department leadership. By handling this situation professionally and through proper channels, I helped ensure the issue was addressed fairly while reinforcing standards of respectful communication and maintaining a safe learning environment.

Finally, my experience as a Teaching Assistant for multiple undergraduate and graduate courses has strengthened my ability to communicate effectively with students at different academic

levels. This experience allows me to anticipate common challenges and provide targeted support that helps students succeed in demanding courses.

Overall, my teaching accomplishments demonstrate effectiveness in instruction, fairness in assessment, strong student support, and professionalism. Through these contributions, I have helped create a learning environment in which students feel supported, treated equitably, and empowered to succeed. These accomplishments align closely with the criteria for the Distinguished Teaching Assistant Award.

Statement of Teaching Philosophy

My teaching philosophy is grounded in my own experience as a learner. Having been a student and continuing to be one as a graduate student, I approach teaching by modeling the type of instruction I value and expect from my own instructors—clear, respectful, and supportive of meaningful learning. I view teaching as a collaborative process in which the instructor's role is to guide students toward clarity, confidence, and independent thinking. As a Teaching Assistant, I strive to create learning environments that are inclusive, transparent, and intellectually engaging, where students from diverse academic backgrounds can succeed.

A core value that shapes my teaching is clarity. As an international student, I have experienced the challenges that can arise when students' prior education is shaped by different academic systems. These experiences have made me especially aware that many students enter technical courses with anxiety or uncertainty about whether they have the required background knowledge to succeed. I believe that clarity in explanation, expectations, and assessment helps reduce this anxiety and allows students to focus on learning rather than fear of failure. In my teaching, I deliberately begin by reviewing foundational concepts and building up step by step to ensure students have the necessary background before introducing more advanced material. I emphasize clear definitions and explicit connections between concepts, and I aim to explain not only how a method works, but also why it works, helping students build a conceptual framework they can transfer to new problems.

Another key aspect of my teaching philosophy is learning through feedback. I view assessment not merely as a way to measure performance, but as an opportunity for growth. Constructive and timely feedback helps students identify misconceptions, reflect on their problem-solving approaches, and improve in future work. I believe students learn best when feedback is clear, specific, and aligned with transparent grading criteria. By understanding where they went wrong and how to improve, students are empowered to take ownership of their learning.

I also believe that professionalism is an essential component of teaching and learning. The classroom is not only a space for acquiring knowledge, but also a place where students learn how to communicate respectfully, engage with feedback, and navigate academic expectations. I strive to model professionalism, ethical conduct, and respectful communication in all interactions. By setting clear boundaries while remaining approachable, I aim to demonstrate the standards expected in academic and professional environments. This helps students understand that respectful dialogue and accountability are integral to learning.

A further goal of my teaching is to foster student confidence and independence. I aim to guide students toward becoming active problem-solvers rather than passive recipients of information. When students ask questions, I encourage them to articulate their thinking and reflect on their reasoning. By gradually shifting responsibility to the student, I help them build confidence in their own abilities. I believe that confidence grows when students experience progress, understand their mistakes, and realize that struggling with material is a normal part of learning.

Finally, I see teaching as a continuous learning process for myself. I regularly reflect on student questions, common difficulties, and feedback to improve my instructional approach. Teaching

has strengthened my ability to communicate complex ideas clearly, listen carefully to students' needs, and adapt my methods to different learning contexts. These skills are essential not only for effective teaching, but also for mentorship and leadership in academia.

In summary, my teaching philosophy centers on clarity, constructive feedback, professionalism, and student empowerment. I strive to create learning environments where students feel respected, supported, and capable of mastering challenging material. Through thoughtful instruction and reflective teaching practices, my goal is to help students develop both academic competence and confidence that will serve them throughout their educational and professional journeys.

Comments

Please identify what you perceive to be the strengths and weaknesses of this teaching assistant and course. (maximum 5000 characters)

Comments
N/A, my lab TA was Aaron so I didn't really get a chance to interact with Maryam.
Great teacher and easy to understand.
Super helpful during labs. I can't think of any weaknesses.
Good at explaining concepts in discussion
TA Ghamarchehreh was readibly available to answer questions about course content, homework assignments, and general coding principles both in discussion/lab and in office hours + by email.
Caring TA
Wonderful TA who really helped my learning.
She was very communicative and she helped out as much as she can. She would respond to emails in a timely manner and was present for help.
super approachable and helpful
Always available to answer and clarify questions and offer advice.
She was a very knowledgeable intelligent and patient TA. I think she was very helpful and caring about students that came to lab and discussion. No weaknesses I can really observe, sometimes the way she explained things a little fast made me confused.
N/A
Maryam was so nice and cared deeply about student learning. She was incredibly helpful and patient with me whenever I had any questions. I would recommend her every time.

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C111/211 - Machine Learning and Artificial Intelligence for Civil Engineering
Instructor: Prof. Enrique López Droguett, PhD.
Teacher Assistants: Maryam Ghamarchehreh
Winter 2026 - UCLA

HOMEWORK 1

Required deliverables: your submitted work **must** follow the naming convention listed below.

- *HW1.py*
- *Report.pdf*
- *All python scripts, data files and final report should be in a zip folder and uploaded on BruinLearn as HW#-FirstName-LastName.zip, for all homework submissions.*

Note: The report should include a ***brief*** description of the solution you developed for the task, in addition to any numerical/graphical result (maximum of 2 pages unless otherwise requested). Please follow the template on BruinLearn. Use good code practices, commenting and taking care of maintaining an organized console output.

Task 1: Beam Analysis (50 pts)

Equation (1) shows the displacement function of the beam presented in Figure 1.

$$y(x) = \frac{w_0}{120EIL}\left(-x^5 + 2L^2x^3 - L^4x\right) \quad (1)$$

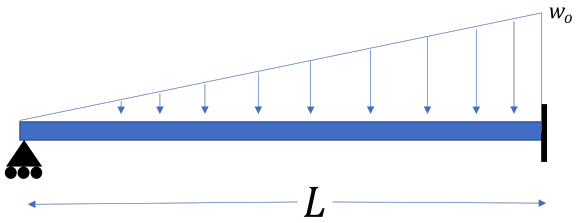


Figure 1: Beam with linearly distributed load.

where E is the modulus of elasticity of the beam, and I is the moment of inertia. Using this equation, write a function for each of the following terms that, given x a single number x, will return the value listed.

a) `displacement(x)` returns the value: $y(x)$
b) `slope(x)` returns the value: $\theta(x) = \frac{dy}{dx}$
c) `moment(x)` returns the value: $M(x) = EI\frac{d^2y}{dx^2}$
d) `shear(x)` returns the value: $V(x) = EI\frac{d^3y}{dx^3}$
e) `loading(x)` returns the value: $w(x) = -EI\frac{d^4y}{dx^4}$

Use the following parameters: $L = 500cm$, $E = \frac{40000kN}{cm^2}$, $I = 30000cm^4$, $w_0 = 3.5\frac{kN}{cm}$ and $\Delta x = 2cm$.

Task 2: Surface Areas in the Solar System (50 pts)

Planets are not perfect spheres. They are better described as oblate spheroids, such as the one shown in Figure 2.

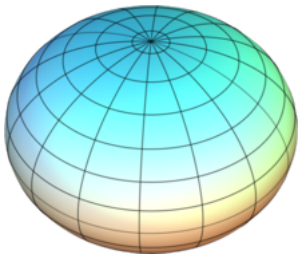


Figure 2: Oblate spheroid. It resembles a beach ball that has been compressed in one axis.

Whereas the surface area of a perfect sphere of radius r is easily computed as $A(r) = 4\pi r^2$, the area of an oblate spheroid is more difficult to compute. The reason is that the description of the object now depends on an equatorial radius r_1 and a polar radius r_2, instead of only one radius r. The formula for the area of an oblate spheroid is given in equation (2):

$$A(r_1, r_2) = 2\pi\left(r_1^2 + \frac{r_2^2}{\sin\gamma} ln\left(\frac{\cos\gamma}{1-\sin\gamma}\right)\right) \quad (2)$$

where $\gamma = arccos\ \frac{r_2}{r_1}$.

Table 1 shows the equatorial and polar radius of all the planets in the solar system. Write functions to compute the True and Approximate surface area of the planet, given the equatorial radius r_1 and the polar radius r_2. The true surface area is computed using equation (2). The approximate surface area is computed using the area of a perfect sphere, where the radius r is the average between r_1 and r_2.

Your functions should have the signatures `true_surface_area(r1,r2)` and `approximate_surface_area(r1, r2)`, and each return a single number. Use your functions to fill in Table 1, and supply the completed table in your report.

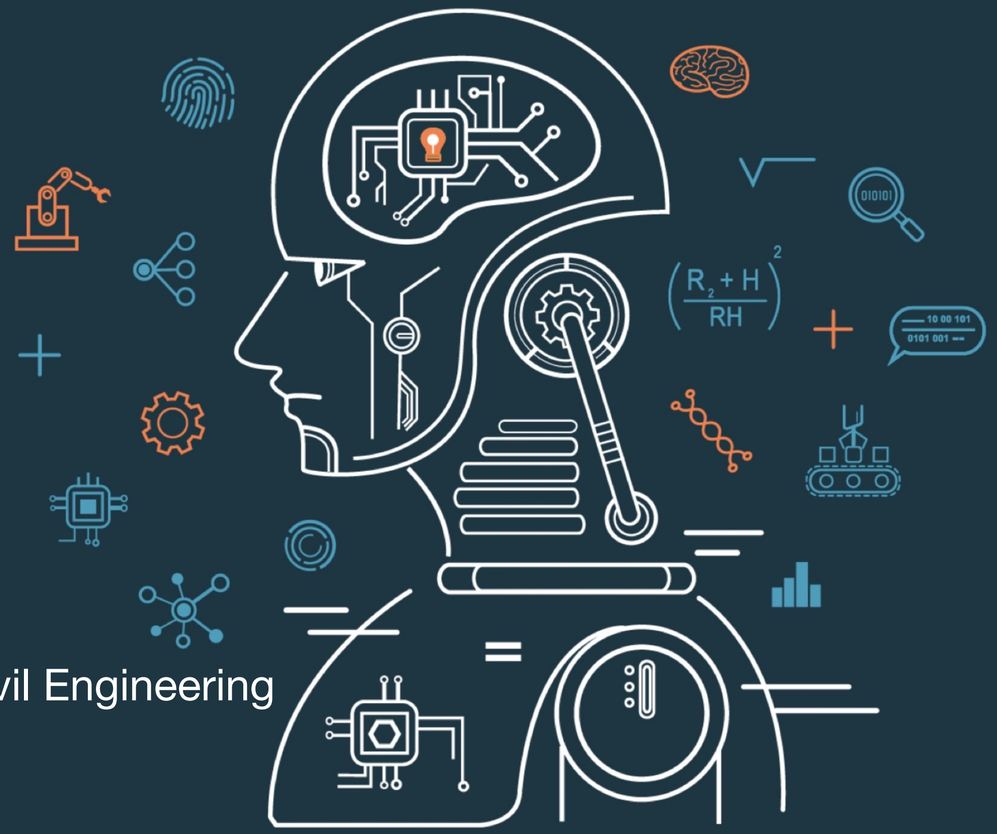
Table 1: Equatorial and Polar radius for the solar system planets. Source: https://nssdc.gsfc.nasa.gov/planetary/factsheet/

Planet	Equatorial Radius r_1 [km]	Polar Radius r_2 [km]	True Surface Area [km^2]	Approximate Surface Area [km^2]
Mercury	2440.5	2438.3	-	-
Venus	6051.8	6051.8	-	-
Earth	6378.1	6371	-	-
Mars	3396.2	3376.2	-	-
Jupyter	71492	66854	-	-
Saturn	60268	54364	-	-
Uranus	25559	24973	-	-
Neptune	24764	24341	-	-

UCLA Samueli School of Engineering

Discussion 1

Python Tutorial



C111/C211- Machine Learning and Artificial Intelligence for Civil Engineering
Winter 2026
UCLA

C111/C222

Teaching Assistant: Maryam Ghamarchehreh
Email: mghamarchehreh@g.ucla.edu

Discussion Section:
Friday, 8:00–9:50 AM
WGYOUNG, Room 4216

Office Hours:
Friday, 10:00 AM–12:00 PM
Boelter Hall, Room 4686

C111/C222

Grading Breakdown:

- 65% Homework
- 35% Final Project

Homework:

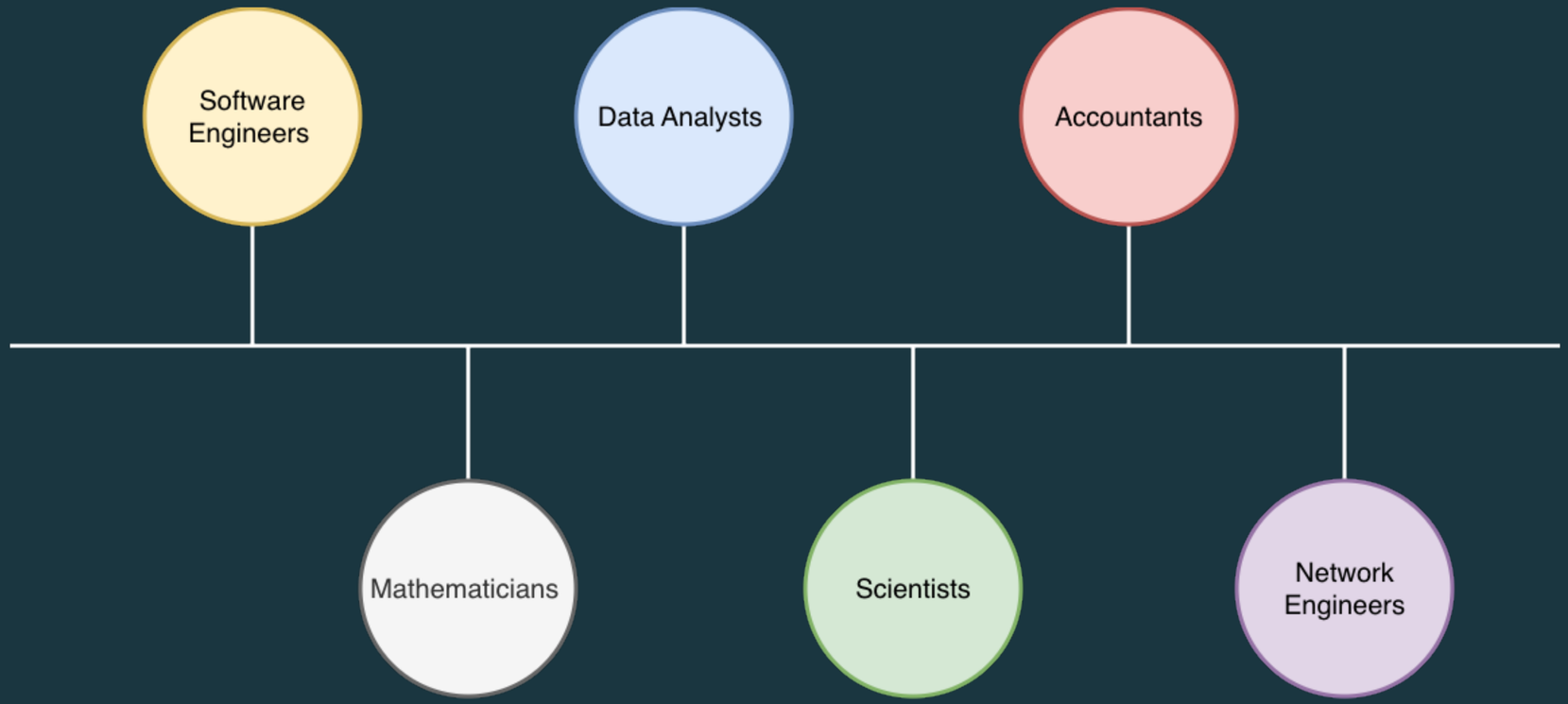
- There are **7 assignments** in total.
- Assignments will be released **every Friday at 7:00 AM**.
- You have **one week** to submit each assignment.
- An additional 2-day grace period is provided with no penalty.

Final Project:

- The final project will be released in **Week 6**.
- You will have **5 weeks** to complete and submit it.

Why Python ?

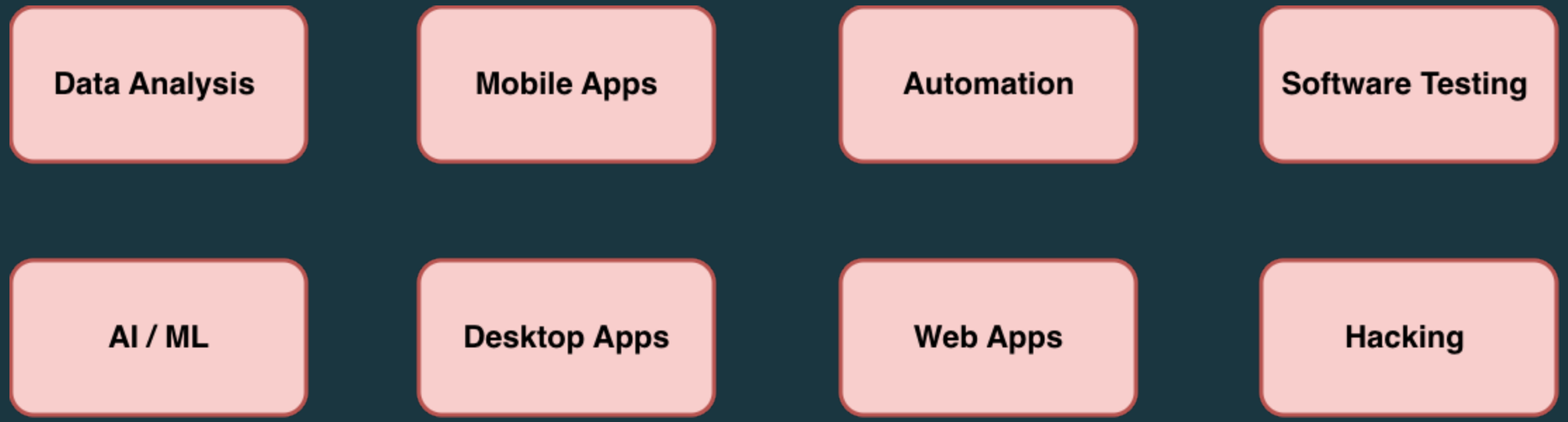
1. Python is the world’s fastest growing programming language.



Why Python ?

2. You can solve complex problems in less time with fewer lines of code.

3. Python is a multi-purpose language.



Why Python ?

4. Python is a high-level language.

5. Python is cross-platform; you can build and run Python applications on Windows, macOS, and Linux.

6. Python has a large ecosystem of libraries, frameworks and tools.



How to write Python Code ?

Editor







IDE
Integrated Development Environment



Notebook-based Environments



First Python Program

What is an expression?

“ An expression is a piece of code that produces a value.”

```
x = "Hello World!"
```

What is a syntax error?

“ A syntax error is a kind of error that is due to bad syntax or bad grammar in the code.”

```
x = "Hello World!"
print x
```

```
  Cell In[7], line 2
    print x
    ^
SyntaxError: Missing parentheses in call to 'print'. Did you mean print(...)?
```

First Python Program

What does a linter do?

“ A linter is a tool that checks our code for potential errors mostly in the category of syntactical errors.”

```
2+
```

```
  Cell In[3], line 1
    2+
      ^
SyntaxError: invalid syntax
```

Variables

We use variable to store data in computer's memory.

What are the primitive types in Python?

- Numbers (integer, float, and complex numbers)
- Booleans
- Strings

```
students_count = 1000   #Integer
rating = 4.99    #Float
is_published = True    #Boolean
course_name = "Python Programming"    #String
```

Strings

Whenever we work with text (string), we should surround your text with single quotes (‘…’) or double quotes (“…”).
We can also use triple quotes (“““ … ”””) to format a long string.

```
course_name = "Python Programming"
message = """
My name is Maryam, and I am the TA for C111 in Winter 2026.
We have our discussion classes on Fridays from 8:00 to 9:50 AM ...
"""
```

Strings

There is another built-in function in Python called “Len” to get the length of a string (number of characters in that string).

```
course = "Python Programming"
len(course)
```

```
18
```

If you want to access a specific character in the string, you use the square bracket notation and enter the index of that character.

```
course = "Python Programming"
print(course[0])
```

```
P
```

Note: In Python, similar to many other programming languages, strings are zero index which mean the index of the first character is zero.

Strings

We can also use negative indices. While the first character in a string has index 0, index –1 refers to the last character of the string.

```
course = "Python Programming"
print(course[-1])
```

```
g
```

We can also slide the string,

```
course = "Python Programming"
print(course[0:3])
print(course[0:])
print(course[:3])
print(course[:])
```

```
Pyt
Python Programming
Pyt
Python Programming
```

Escape Sequences

Assume we have a string that contains a double quote or single quote inside the text. How can we write it in Python?

We can use a backslash before the double or single quote to escape it:

```
course_1 = "Python \" Programming"
course_2= "Python \' Programming"
print(course_1)
print(course_2)
```

```
Python " Programming
Python ' Programming
```

❓ If the text contains a backslash, how can we write it in Python?

Escape Sequences

If we want to print part of a string on a new line, we can use \n.

```
course = " Python \n Programming"
print(course)
```

```
 Python
 Programming
```

Formatted Strings

Here we have two variables, first and last. When we want to see the full name on the console :
1. We can define a new variable, use concatenation and have a new string.

```
first = "Maryam"
last = "Ghamarchehreh"
full = first + " " + last
print (full)
```

```
Maryam Ghamarchehreh
```

2. Or we can use formatted strings :

```
first = "Maryam"
last = "Ghamarchehreh"
full = f"{first} {last}"
print (full)
```

```
Maryam Ghamarchehreh
```

Formatted Strings

We can put any valid expressions between curly braces :

```
first = "Maryam"
last = "Ghamarchehreh"
full = f"{len(first)} {2+2}"
print (full)
```

```
6 4
```

String Methods

Everything in Python is an object. Objects have functions we called methods that we can access using the dot notation. Let's take a look at few of these methods :

```
course = "  Python programming  "
print(course.upper()) #To convert the string to uppercase
print(course.lower()) #To convert the string to lowercase
print(course.title()) #Capitilize the first letter of every word
print(course.strip()) #Remove white spaces from the beginning and end of the string
print(course.find("pro")) #To find the index of the argument
print(course.replace("p","j")) #Replace a character or a sequence of characters with the argument
print("pro" in course) #Check for the existance of a character or a sequence of  characters in the string
print("swift" not in course) #To check if a character or a sequence of  characters don't exist in the string
```

```
  PYTHON PROGRAMMING
  python programming
  Python Programming
Python programming
9
  Python jrogramming
True
True
```

Numbers

In Python, we have 3 different types of numbers:
1. Integers 2. Floats 3. Complex Numbers

```
x = 1
y = 1.1
z = 1 + 2j
```

For all these numbers, we have standard arithmetic operations:

```
print(10+3)  #Addition
print(10-3)  #Subtraction
print(10*3)  #Multiplication
print(10/3)  #Division (Result can be a floating numebr)
print(10//3) #Division (Result is only an integer)
print(10%3)  #Modulus (Remainder of the division)
print(10**3)  #Exponent (Left to the power of right)
```

```
13
7
30
3.3333333333333335
3
1
1000
```

Numbers

For all those operators, we have a special operator called augmented assignment operator.
Assume we have x = 10, we want to increment x by 3.

```
x = 10
x = x + 3
x += 3
```

Working with Numbers

We have only a handful of built-in functions to work with numbers :

```
print(round (2.9))
print(abs (-2.9))
```

```
3
2.9
```

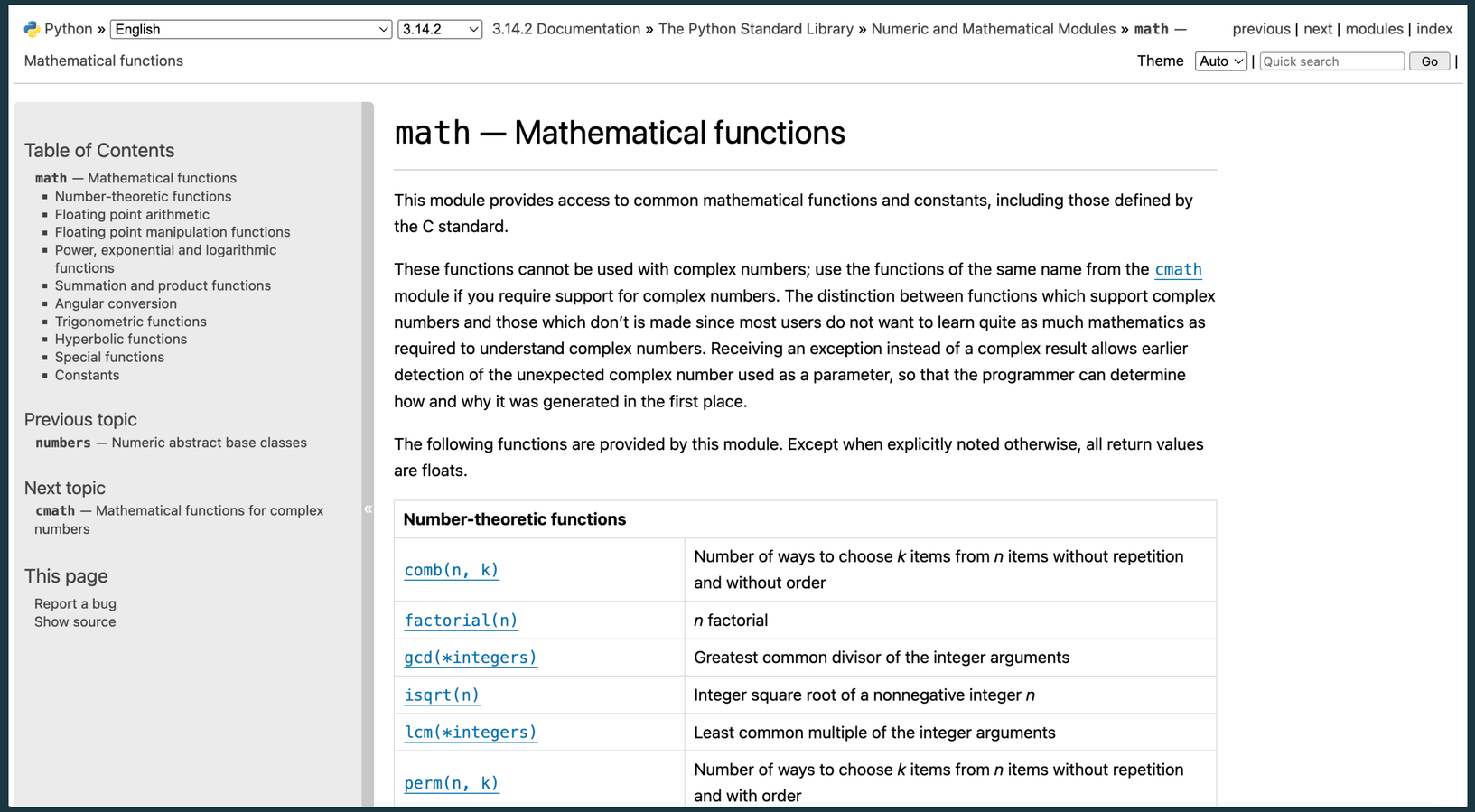
If we want to write a program that involves complex mathematical calculations, we need to use the math module. In Python, we have “Math Module” which includes some mathematical functions for working with numbers. We need to import this module so we can use it.

```
import math
print (math.ceil(2.9))
```

```
3
```

Working with Numbers

To find the complete list of functions in math module , check https://docs.python.org/3/library/math.html



Python » English 3.14.2 3.14.2 Documentation » The Python Standard Library » Numeric and Mathematical Modules » **math** — previous | next | modules | index

Mathematical functions Theme Auto | Quick search Go |

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- Floating point arithmetic
- Floating point manipulation functions
- Power, exponential and logarithmic functions
- Summation and product functions
- Angular conversion
- Trigonometric functions
- Hyperbolic functions
- Special functions
- Constants

Previous topic

numbers — Numeric abstract base classes

Next topic

cmath — Mathematical functions for complex numbers

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`math` — Mathematical functions

This module provides access to common mathematical functions and constants, including those defined by the C standard.

These functions cannot be used with complex numbers; use the functions of the same name from the `cmath` module if you require support for complex numbers. The distinction between functions which support complex numbers and those which don't is made since most users do not want to learn quite as much mathematics as required to understand complex numbers. Receiving an exception instead of a complex result allows earlier detection of the unexpected complex number used as a parameter, so that the programmer can determine how and why it was generated in the first place.

The following functions are provided by this module. Except when explicitly noted otherwise, all return values are floats.

Number-theoretic functions	
`comb(n, k)`	Number of ways to choose *k* items from *n* items without repetition and without order
`factorial(n)`	*n* factorial
`gcd(*integers)`	Greatest common divisor of the integer arguments
`isqrt(n)`	Integer square root of a nonnegative integer *n*
`lcm(*integers)`	Least common multiple of the integer arguments
`perm(n, k)`	Number of ways to choose *k* items from *n* items without repetition and with order

Type Conversion

Another useful function in Python is “input”.

```
x = input ("x: ") #This asks user to enter x value
print (type(x)) # x is stored as a string
X = int(x) # To apply mathematical operations on x, we need to convert it to integer or float
y = X +3
print (f"x:{X},y:{y}")
```

```
x:  1
<class 'str'>
x:1,y:4
```

Comparison Operators

```
10>3
```

```
True
```

```
10>=3
```

```
True
```

```
10==10
```

```
True
```

```
10<3
```

```
False
```

```
10=="10"
```

```
False
```

```
10!="10"
```

```
True
```

Comparison Operators

We can also use comparison operators for strings as well.

```
"bag" > "apple" #When we sort these two words, "bag" comes after "apple"
```

```
True
```

```
"bag" == "BAG"
```

```
False
```

Conditional Statements

```
temperature = 35
if temperature > 30:
    print("It's warm")
print("Done")
```

```
It's warm
Done
```

```
temperature = 15
if temperature > 30:
    print("It's warm")
print("Done")
```

```
Done
```

```
temperature = 15
if temperature > 30:
    print("It's warm")
elif temperature > 20:           # It is short for "else if"
    print("It's fine")
else:
    print("It's cold")
print("Done")
```

```
It's cold
Done
```

Logical Operators

In Python, we have three logical operators to model more complex conditions:

- and
- or
- not

```
high_income = False
good_credit = True
if high_income and good_credit:
    print("Eligible")
else:
    print("Not Eligible")
```

```
Not Eligible
```

```
high_income = False
good_credit = True
if high_income or good_credit:
    print("Eligible")
else:
    print("Not Eligible")
```

```
Eligible
```

Logical Operators

```
high_income = False
good_credit = True
student = True
if not student:
    print("Eligible")
else:
    print("Not Eligible")
```

```
Not Eligible
```

```
high_income = False
good_credit = True
student = False
if (high_income or good_credit) and not student:
    print("Eligible")
else:
    print("Not Eligible")
```

```
Eligible
```

“For” Loops

For loop is used to create repetition.

```
for number in range(3):
   #Write all the statements that need to be repeated 3 times
    print("Attempt",number,(number)*".")
```

```
Attempt 0
Attempt 1 .
Attempt 2 ..
```

```
for number in range(3):
   #Write all the statements that need to be repeated 3 times
    print("Attempt",number+1,(number+1)*".")
```

```
Attempt 1 .
Attempt 2 ..
Attempt 3 ...
```

```
for number in range(1,10,2):    #range(start,end,step_size)
   #Write all the statements that need to be repeated 3 times
    print("Attempt",number,(number)*".")
```

```
Attempt 1 .
Attempt 3 ...
Attempt 5 .....
Attempt 7 .......
Attempt 9 .........
```

For … Else

```
successful = True
for number in range(3):
    print("Attempt")
    if successful:
        print("Successful")
        break
else:
    print("Attempted 3 times and failed")
```

```
Attempt
Successful
```

```
successful = False
for number in range(3):
    print("Attempt")
    if successful:
        print("Successful")
        break
else:
    print("Attempted 3 times and failed")
```

```
Attempt
Attempt
Attempt
Attempted 3 times and failed
```

Nested Loops

We can put one loop inside of another loop which is called “ Nested Loop”.

```
for x in range(5):   #Outter loop
    for y in range(3):   #Inner loop
        print(f"({x},{y})")
```

```
(0,0)
(0,1)
(0,2)
(1,0)
(1,1)
(1,2)
(2,0)
(2,1)
(2,2)
(3,0)
(3,1)
(3,2)
(4,0)
(4,1)
(4,2)
```

Iterables

As you can see, in addition to string, integer, float, and boolean, we have some complex classes such as range which is iterable (we can iterate a process over it).

Range objects are not the only iterable function is Python, Strings are also iterable.

```
for x in "Python":    # In this iteration, x will hold one character in the string
    print(x)
```

```
P
y
t
h
o
n
```

Iterables

Another complex class called list which is used to store a list of objects.

```
for x in [1,2,3,4]:
    print(x)
```

```
1
2
3
4
```

While Loops

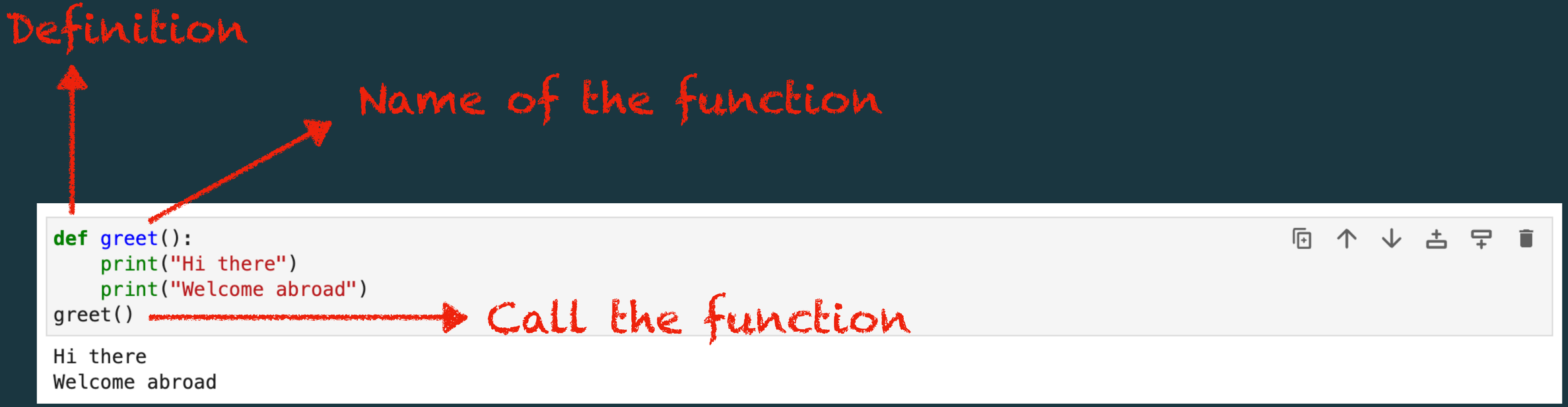
We use While loops to repeat something as long as a condition is true.

```
number = 100
while number > 0:
    print(number)
    number //= 2
```

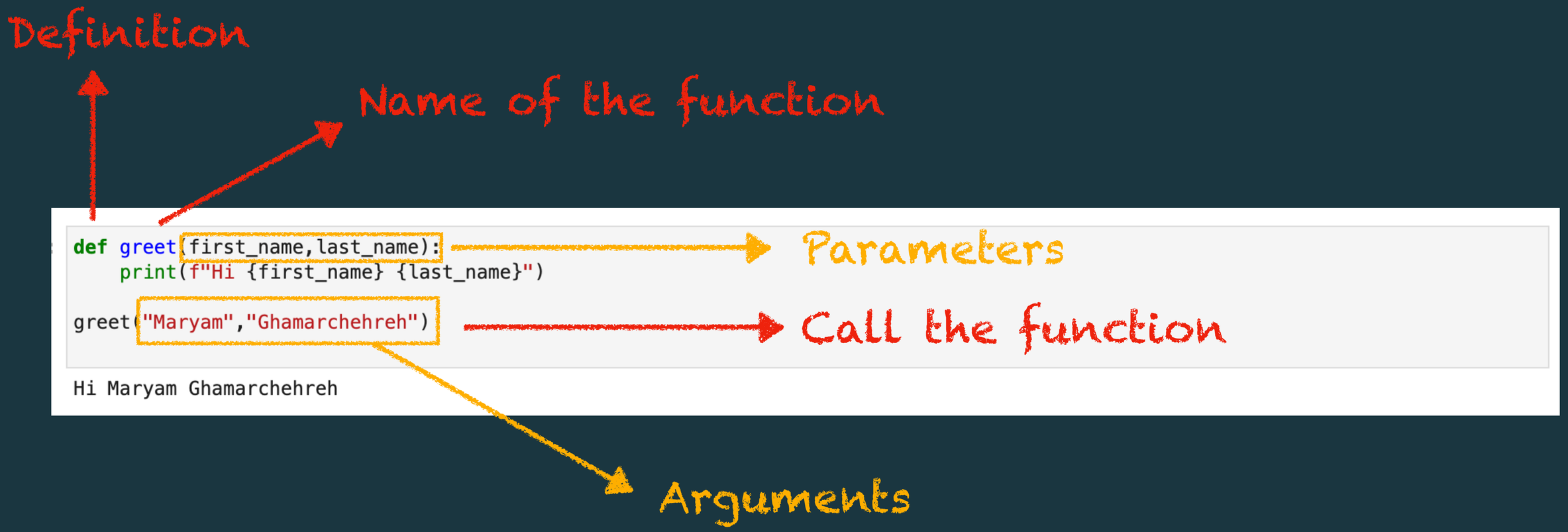
```
100
50
25
12
6
3
1
```

Functions

Now we want to learn how to write our own function



Arguments

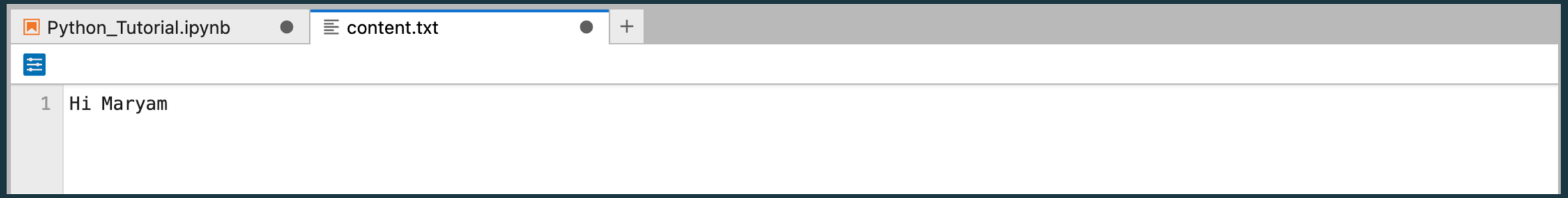

Definition
Name of the function
def greet(first_name,last_name):
print(f"Hi {first_name} {last_name}")
Parameters
greet("Maryam","Ghamarchehreh")
Call the function
Hi Maryam Ghamarchehreh
Arguments

Types of functions

In programming, we have two types of functions:
1. Function that performs a task
2. Function that calculates and returns a value

```python
def get_greeting(name):   #This function simply return a value
    return f"Hi {name}"

message = get_greeting("Maryam")
file=open("content.txt","w")
file.write(message)
file.close()
```



Args, wait, what?

```
def multiply(x,y):
    return x * y

multiply(2,3,4,5)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[139], line 4
      1 def multiply(x,y):
      2     return x * y
----> 4 multiply(2,3,4,5)

TypeError: multiply() takes 2 positional arguments but 4 were given
```

How we can fix this?

```
def multiply(*numbers):   #In this way numbers is a tuple which is an iterable collection of objects like
                          # a list with one differrence, we can not modify objects
    total=1
    for number in numbers:
        total*=number
    return total

multiply(2,3,4,5)
```

```
120
```

Lambda Function

Lambda Function is a small anonymous function for a one time use. It takes any number of arguments, but only have 1 expression. It helps keep the namespace clean and is useful with higher order functions “sort()”, “map()”, ”filter()”, “reduce()”.

```
double = lambda x: x*2
add = lambda x,y: x+y
max_value = lambda x,y: x if x>y else y
full_name = lambda first, last: first + " " +last
```

```
print(double(4))
print(add(2,3))
print(max_value(5,6))
print(full_name("Maryam","Ghch"))
```

```
8
5
6
Maryam Ghch
```

Files

HW1 - Raphael - Pilleul.zip 19.2 KB

Submitted: Jan 16 at 12:18pm late

Days 0.22

Due: Fri Jan 16, 2026 7:00am

Submitted Files: (click to load)

HW1 - Raphael - Pilleul.zip

Assessment

Grade out of 100

98

View Rubric

Some Rubric

Criteria	Ratings
Task 1	**No details** **Comments** Results need to be reported for x=0:2:500 48 / 50 pts
Task 2	**Full Marks** 50 / 50 pts
	Total Points: 98

Support

```
#!/usr/bin/env python3
"""
================================================================================
C111/222 - Machine Learning and Artificial Intelligence for Civil Engineering
Homework 3: Ball Bearing Health State Diagnosis Using Unsupervised Learning

Author:     Alejandro Angel
UCLA ID: 006377548
Winter 2026 - UCLA

Description:
    This script performs health state diagnosis of ball bearings using
    unsupervised learning techniques (K-Means, DBSCAN) with and without PCA.

Usage:
    python HW3.py

    Or import as module:
    from HW3 import BallBearingAnalysis
    analysis = BallBearingAnalysis('BallBearings.csv')
    analysis.run_full_analysis()
================================================================================
"""

# ============================================================================
# IMPORTS
# ============================================================================
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from mpl_toolkits.mplot3d import Axes3D
import warnings
import os
from datetime import datetime

# Machine Learning libraries
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

Assessment

Grade out of 100

90

View Rubric

Some Rubric (2)

Criteria	Ratings
Task 1. Data Preprocessing	**Full Marks** 20 / 20 pts
Task 1. K-Means and DBSCAN Clustering	**No details** **Comments** Please generate two three-dimensional plots using two different combinations of input variables (−5) 35 / 40 pts
Task 1. PCA Clustering	**No details** **Comments** Please determine the clusters and create plots for every pair of components for each method (−5) 35 / 40 pts

Total Points: 90

Comments for this Attempt

Support

Files

HW1-Daniel-Tritasavit.zip 104 KB

Submitted: Jan 16 at 4:55pm late

Days 0.41

Due: Fri Jan 16, 2026 7:00am

Submitted Files: (click to load)

HW1-Daniel-Tritasavit.zip

Assessment

Grade out of 100

93

View Rubric

Some Rubric

Criteria	Ratings
Task 1	**No details** **Comments** Results need to be reported for x=0:2:500 48 / 50 pts
Task 2	**No details** **Comments** You need to define a dictionary containing the planets and their radii, and then call your functions using this dictionary. 45 / 50 pts

Support

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name

Briefly describe how you know the nominee. (50 words max)

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

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Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

The B. John Garrick Institute for the Risk Sciences
UCLA ENGINEERING

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Box 159510
Los Angeles, CA 90095
Tel: 310.825.5534
Fax: 310.206.7353

Letter of Support for Distinguished Teaching Award

To the Selection Committee,

I am writing to offer my strongest support for Maryam Ghamarchehreh in nomination for the Distinguished Teaching Assistant Award. I have had the pleasure of working directly with Maryam in her role as a Teaching Assistant for C&EE 110: Introduction to Probability and Statistics for Engineers, M20: Introduction to Computer Programming with MATLAB, and C&EE 111/C 211: Machine Learning and Artificial Intelligence for Civil Engineering in the Department of Civil and Environmental Engineering. I can confidently state that she exemplifies the highest standards of teaching excellence, professionalism, and dedication to student learning.

Throughout these courses, which were attended by students from diverse academic backgrounds and varying levels of prior preparation, Maryam demonstrated exceptional commitment to both undergraduate and graduate instruction. She was consistently well prepared for discussion sections, explained complex concepts with clarity and patience, and skillfully adapted her teaching approach to meet the needs of students with different levels of technical experience. Her ability to break down challenging material into accessible and well-structured explanations significantly enhanced student understanding, engagement, and confidence.

Beyond content mastery, Maryam fostered an inclusive and respectful learning environment. She encouraged questions, welcomed differing perspectives, and ensured that all students felt comfortable participating. Students frequently expressed appreciation for her approachability, responsiveness, and genuine investment in student academic success. This supportive atmosphere contributed to increased confidence and independent thinking among students.

She also displayed strong organizational and leadership skills. She handled grading responsibilities with fairness and transparency, provided detailed and constructive feedback, and was highly organized in responding promptly to student emails and holding regular, well-structured office hours and discussion sessions. She also maintained clear and effective communication with both students and instructional staff. Her professionalism and reliability made her an invaluable member of the teaching team.

In summary, Maryam has made a meaningful and lasting impact on the educational experience of our students. Her dedication to teaching, commitment to equity and clarity, and excellence in instructional practice make her exceptionally deserving of the Distinguished Teaching Awards. I offer my strongest recommendation without reservation.

Please feel free to contact me if additional information is needed.

Sincerely,



Enrique López Droguett
Professor, Civil and Environmental Engineering Department
Director, Center for Reliability Science and Engineering